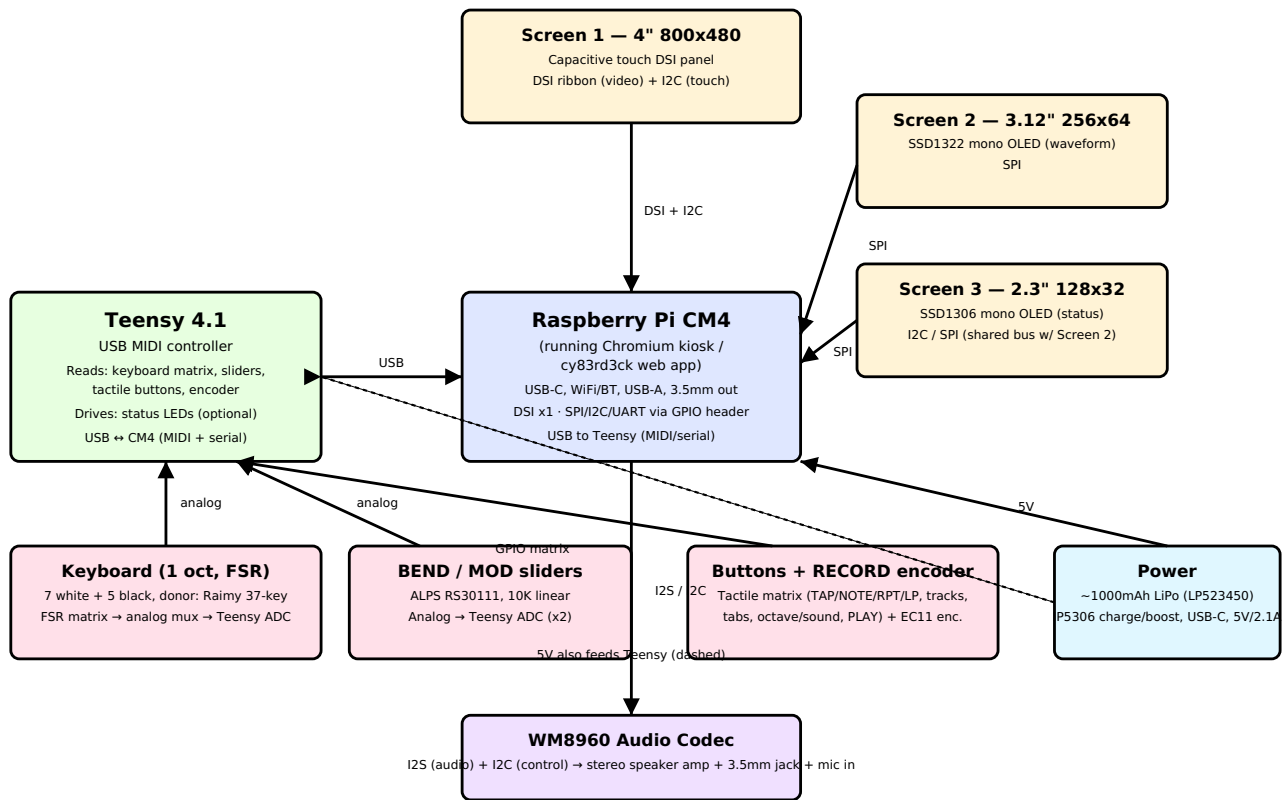


cy83rd3ck — SYSTEM BLOCK / WIRING DIAGRAM



Notes

- CM4 module: 8GB RAM / 16GB eMMC / Wireless variant (CM4108016) — gives Chromium kiosk headroom, onboard storage for app+samples (no SD slot needed), and built-in WiFi/BT.
- CM4 hosts the browser-based cy83rd3ck app and drives Screen 1 (DSI touchscreen) directly via the DSI connector + I2C touch.
- Teensy 4.1 acts as the dedicated input controller: reads the keyboard FSR matrix (via analog mux), BEND/MOD sliders (analog), and the full button/encoder matrix (GPIO), then sends MIDI/serial messages to the CM4 over USB.
- Screens 2 (waveform OLED) and 3 (status OLED) share an SPI bus off the CM4 GPIO header, each on its own CS line.
- Audio: WM8960 codec on I2S + I2C from the CM4 — built-in stereo speaker amp drives the two 16mm speakers, plus headphone amp for the 3.5mm jack and a mic input, all in one small IC (no USB audio dongle needed).
- Power: single LiPo with USB-C charge/boost circuit feeds both CM4 (5V/PD) and Teensy (5V via USB or VIN).
- This is a system-level block diagram — actual GPIO pin assignments and the keyboard/button matrix wiring (rows/columns) would be defined in the PCB schematic (KiCad) as a follow-up step.